

My approach to teaching is centered around my own love for learning. When I am learning something new, especially something I find interesting that peaks my curiosity, I become captivated and hungry to learn more. A good analogy is that feeling you get when you find a good book that you cannot put down. I often want to go beyond just memorizing and understanding the topic; I want to experiment with this new knowledge and create something new.

This feeling of excitement and enthrallment is what I want to arouse in my students as an instructor. I want to spark the students' interests and make them hungry for more. I want to motivate them to not just remember concepts but to imagine how they can apply new topics in a way that interests them and then enable them to create the things they envision. This mindset is what defines my teaching philosophy and what has shaped a lot of my prior experiences in teaching.

1 Teaching Philosophy

I believe that active learning and student engagement are central to effective education. I take inspiration from Bloom's Taxonomy for classifying educational goals (Bloom et al. [1956]) when designing my courses and creating assignments. I design my course content to start at the *Understand* and *Apply* level of the hierarchy, where students initially gain a basic understanding of the topics and how they are applied. I use summative assessments that help students strengthen their understanding of the topics and identify points of confusion, and to act as feedback on my approach to teaching. Examples include in-class worksheets, 60-second essays, and short quizzes. I then create formative assessments that help push the students further up the pyramid to the *Create* and *Evaluate* phases of learning through projects and larger course assignments. Examples of these types of assessments include scaffolded and course-long projects, independent literature reviews, and exams. I believe that students should receive continuous feedback during both the acquisition and application phases of the learning process, with an emphasis on iterating over formative assessments to help students further refine outcomes.

I use a "problem-based learning" approach when designing my course assessments (Hannafin and Land [2000]), where I endeavor to ground course assignments in applications relevant to the student's future career aspirations and encourage "learning by doing" as a philosophy. I like to give students the opportunity to propose and design their own project outcomes based on their interests and what excites them. I encourage students to go beyond just the topics covered in lectures and to explore new solutions, tools, and applications that were not covered in the course but connect back to the topics and themes of the course. In practice, I like to have a mix of simplistic, "build-your-own" projects and projects that allow students to build something while leveraging existing tools (Touretzky [2013]). Examples include writing the A^* search algorithm from scratch to get a better understanding of path

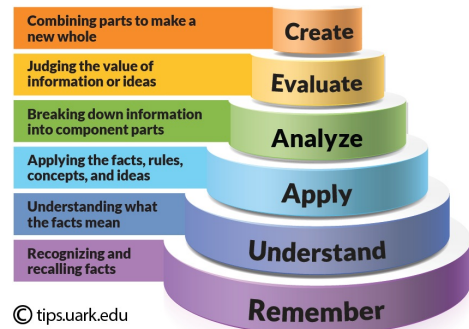


Figure 1: Bloom's taxonomy for educational goals.

planning and then building on top of a well developed implementation of RRT^* to create a solution to an application that interests the student.

When designing my lectures, I use a student-centered learning pathways philosophy (Trefny Center [2025]). I start by making a connection to the topic of discussion with a previous topic or with something that motivates the new topic from applications outside of the classroom. This leads into the knowledge acquisition phase of the lecture. I prefer to use a mix of slides and examples written out on the board. I encourage questions and often integrate various pauses into my lectures to solicit questions from the students. I occasionally pose questions to the class to keep students engaged and to give them an opportunity to think through the ideas and how they connect to one another. As we dive deeper into a topic, I evaluate student outcomes through project-based formative assessments. I believe that this repeated exposure to course content is necessary for concepts to stick. Keeping with the problem-based learning philosophy, I am always trying to connect my lecture content with real-world applications.

2 Teaching Experience

Most of my teaching experience stems from my time teaching the Computer Science for STEM course (Colorado School of Mines catalog: *CSCI 128*). I taught *CSCI 128* for two semesters at the Colorado School of Mines. This course teaches students the fundamentals of computer programming using Python and averages between 50-70 students per section. The course is required for all students at Mines and the majority of the students are freshmen who are not majoring in Computer Science.

Teaching *CSCI 128* has taught me a lot about leading a class and how to give a lecture. The majority of the course content was provided to me, but I still tailored the lecture slides to fit with my student-centered learning philosophy. I also had the opportunity to write exam questions for the course and to discuss how to improve the course with other instructors. Several very experienced instructors teach the course each semester and I learned a lot about teaching pedagogy and how to run a university-level course while working with them.

I have also completed the College Teaching Certificate Program, led by the Trefny Innovative Instruction Center at the Colorado School of Mines. This program is designed to help early-career academics prepare to teach at the university level by studying pedagogy and learner-centered practices. Throughout the program, I designed an upper-level undergraduate course to introduce students to robot autonomy. I received a lot of constructive feedback on both the course and my in-class teaching from *CSCI 128*.

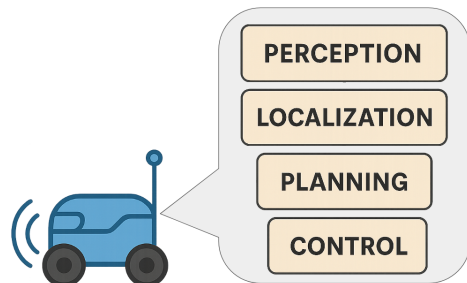


Figure 2: Course logo for Introduction to Robot Autonomy.

3 Courses I Can Offer

I would love to offer the course that I developed through the College Teaching Certificate Program on Robot Autonomy. This course would introduce students to the common, major components that typically make up the robot autonomy stack and is geared towards upper-level undergraduates. I could readily expand the course for graduate students as well. I would also like to develop a graduate-level course on Operations Research for Computer

Scientists that would introduce students to common mathematical optimization solvers and investigate the algorithms that make these solvers work. I would also like to lead a seminar course on applied artificial intelligence for robotics, where we read through a curated list of seminal papers on this topic and discuss what made these works so impactful.

I am able to teach several core Computer Science courses as well. I feel quite comfortable with courses that are heavy in theory and mathematics as well as more applied courses. Some example courses that come to mind include Algorithms, Data Structures, Discrete Mathematics, and Software Engineering. Possible elective courses include Artificial Intelligence and Embedded Systems.

4 Mentoring and Advising Experience

I have had the privilege of mentoring several graduate and undergraduate students. While working in my advisor's research lab, I have mentored four graduate students in their own research and have lead eight Mines Undergraduate Research Fellowship (MURF) scholars in various research projects. These students have either first-authored or co-authored seven different research publications while working with me. Beyond research, I have had the pleasure of providing academic and career advice to these students and many have gone on to work or continue their studies at various corporations and institutions, including: the University of Colorado Boulder, Colorado School of Mines, Northrop Grumman, Trimble, Anduril, and Outside Analytics.

I have also advised several students from my *CSCI 128* sections. Common topics include how to succeed in the course, how to get involved in research, and what elective courses best match their interests. I have also had to have some difficult discussions with these students, including cheating and determining when to drop the course. This experience, both the good and the bad, have taught me a lot about how to run a class and how the students look up to me as their instructor.

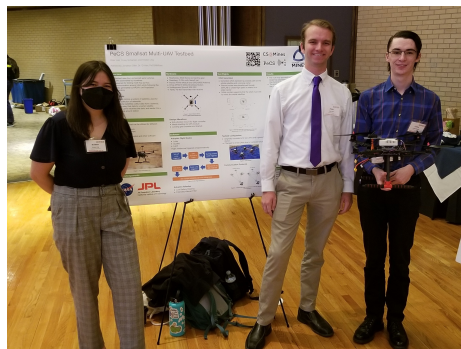


Figure 3: MURF scholars presenting their drone project.

Influential Literature

Benjamin S. Bloom, Max D. Engelhart, Edward J. Furst, Walker H. Hill, and David R. Krathwohl. *Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook I: Cognitive Domain*. David McKay Company, New York, 1956.

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